



Experiment-2

Name of the experiment: Determination of uniaxial compressive strength of rock samples

INTRODUCTION

One of the important engineering properties of the rock, which plays an essential role in designing various geotechnical applications such as underground mines, tunnel and dam is Uniaxial Compressive Strength (UCS) of the core specimen. Compressive strength is a key value for design of underground structures. The determination of compressive strength involves the use of a cylindrical specimen of rock loaded axially between platens in a testing machine. The stress value at failure is defined as the compressive strength of the specimen. There is no load along an axis perpendicular to the loading axis. The loading condition simulates the condition of pillars supporting the mine roof in case of underground mines. The Uniaxial compressive strength depends on friction between platens and end surfaces, specimen geometry, rate of loading and water content. If the friction between loading plate and the rock specimen is more, the specimen will fail in shear, whereas in case of a smooth contact between the loading plate and the specimen, the specimen will fail in tension. The compressive strength of rocks usually increases with increase in the rate of loading of specimen. The strength of the specimen usually decreases with increase in their size and when the ratio of length to diameter increases the compressive strength will be decreases. Dry specimens have high compressive strength and saturated specimens have low values.

SCOPE

This method of test is intended to measure the uniaxial compressive strength of a rock sample in the form of specimens of regular geometry. The test is mainly intended for strength classification and characterization of intact rock.

APPARATUS

(a) A suitable machine (Figure 1) shall be used for applying and measuring axial load to the specimen. It shall be of sufficient capacity and capable of applying load at a rate conforming to the requirements.



Fig. 1: Universal Testing Machine

- (b) Equipments for measuring dimensions of the rock sample as shown in Figure 2.
(c) Equipments for measuring weight of the rock sample as shown in Figure 2.



Fig. 2: Digital Vernier and Electronic balance

PROCEDURE

- (a) Test specimens shall be right circular cylinders having a height to diameter ratio of 2.5-3.0 and a diameter preferably of not less than NX core size, approximately 54 mm. The diameter of the specimen should be related to the size of the largest grain in the rock by the ratio of at least 10:1.
- (b) The ends of the specimen shall be flat to 0.02 mm and shall not depart from perpendicularity to the axis of the specimen by more than 0.001 radian (about 3.5 min) or 0.05 mm in 50 mm.
- (c) The sides of the specimen shall be smooth and free of abrupt irregularities and straight to within 0.3 mm over the full length of the specimen.
- (d) The use of capping materials or end surface treatments other than machining is not permitted.
- (e) The diameter of the test specimen shall be measured to the nearest 0.1 mm by averaging three diameters measured at right angles to each other at about the upper-height, the mid-height and the lower height of the specimen. The average diameter shall be used for calculating the cross-sectional area. The height of the specimen shall be determined to the nearest 1.0 mm.
- (f) Samples shall be stored, for no longer than 30 days, in such a way as to preserve the natural water content, as far as possible, and tested in that condition.
- (g) Load on the specimen shall be applied continuously at a constant stress rate such that failure will occur within 5-10min of loading, alternatively the stress rate shall be within the limits of 0.5-1.0 MPa/sec.
- (j) The number of specimens tested should be determined from practical considerations but at least five are preferred.

The uniaxial compressive strength (σ_c) of the specimen shall be calculated by dividing the maximum load (P) carried by the specimen during the test, by the original cross-sectional area (A).

$$\sigma_c = \frac{P}{A}$$

Observations:

Sample No.	Rock Type	Diameter			Avg. Diameter (mm)	Length			Avg. Length (mm)	Weight (kg)	Breaking load (tonne)	Total time taken (sec.)	Type of failure
		Top (mm)	Middle (mm)	Bottom (mm)		Top (mm)	Middle (mm)	Bottom (mm)					
<i>Example</i>	Charnockite	54.28	54.28	54.32	54.29	138.3	138.32	138.42	138.35	0.871	28.35	115	Shear
1.													
2.													

Calculations:

Calculate the cross-sectional area of the specimen as follows:

$$A = (\pi D^2) / 4$$

where: A = cross-sectional area, in (mm²)

D = average specimen diameter, in (m)

Calculate the volume of the specimen as follows:

$$V = A \times L$$

where: V = volume, in (m³)

A = cross-sectional area, in (mm²)

L = specimen length, in (m)

Calculate the specimen unit weight as follows:

$$UW = (M \times g) / V$$

where: UW = specimen unit weight, in (N/ m³)

M = specimen mass, in (kg)

W = specimen weight, in (N)

V = volume, (m³)

Calculate the compressive strength in the test specimen from the maximum compressive load on the specimen and the initial computed cross-sectional area as follows:

$$\sigma_c = \frac{P}{A}$$

where: σ_c = compressive strength, in MPa

P = maximum load, in (kN)

A = cross sectional area, in m²



Before



After

Examples of Calculations:

Given:

Average diameter = 54.29mm

Length = 138.35mm.

Mass = 0.871kgs

Maximum load = 28.35 tonnes

Calculation:

$$L/D = 138.35 / 54.29 = 2.55$$

$$A = [\pi \times (54.29)^2] / 4 = 23.15 \times 10^{-4} \text{ m}^2$$

$$V = (23.15 \times 10^{-4}) \times (0.13835) = 3.202 \times 10^{-4} \text{ m}^3$$

$$\text{Density } (\rho) = 0.871 / 3.202 \times 10^{-4} = 2720 \text{ kg/ m}^3$$

$$\text{Unit weight } (\gamma) = \rho \times g = 2720 \times 9.81 = 26683 \text{ N/ m}^3$$

$$\sigma_c = \frac{\text{Breaking load} \times g}{A} = \frac{28.35 \times 10^3 \times 9.81}{23.13 \times 10^{-4}} = \underline{120.08 \text{ MPa}}$$

$$\text{Rate of stress} = (120.08 \text{ MPa}) / 115 = \underline{1.04 \text{ MPa/Sec.}}$$

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.